

Super-Resolution Sea Ice Concentration using Generative AI

Danish: Super-Opløsning af Havis ved brug af Generativ KI

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Motivation & background

The aim of this thesis is to develop a novel framework for achieving super-resolution of sea ice concentration (SIC) fields derived from coarse passive microwave datasets. The primary objective is to enhance these low-resolution (12.5 to 50.0 km grid resolution [1]) brightness temperature (BT) fields from the Advanced Microwave Scanning Radiometer 2 (AMSR-2) to match the spatial granularity of the Danish Meteorological Institute's Automated Sea Ice Products (DMI-ASIP), which delivers high-resolution sea ice maps at 500-meter and 80-meter scale at level 3 and 2 respectively [2].

The enhancement of SIC resolution is critical for improving short-term sea ice forecasting, navigation, and climate monitoring in the rapidly evolving Arctic environment. To address the inherent limitations of passive microwave remote sensing, the approach is to leverage recent advancements in deep generative modeling.

The data for this project is a temporally and spatially co-located AMSR-2 BT and DMI-ASIP dataset, which will be used to demonstrate a significantly outperforming of traditional interpolation techniques in terms of both visual coherence of ice edges, but also stochastic metrics like Continuous Rank Probability Scores. This study illustrates the feasibility and robustness of applying advanced generative models to remote sensing of sea ice, providing a valuable tool for operational ice monitoring and pave the way towards a 5 km Climate Data Record (CDR) exploiting the upcoming Copernicus Imaging Microwave Radiometer (CIMR) [3].

References

- [1] N. Ivanova, L. T. Pedersen, R. T. Tonboe, S. Kern, G. Heygster, T. Lavergne, A. M. Sørensen, R. Saldo, G. Dybkjær, L. Brucker, and M. Shokr, "Satellite passive microwave sea ice concentration data set inter-comparison: Closed maritime basins," *The Cryosphere*, vol. 9, no. 1, pp. 73–89, 2015.
- [2] T. Wulf, J. Buus-Hinkler, S. Singha, N. Dasgupta, A. Athanasiadis, and M. B. Kreiner, "A decade of sea ice concentration retrievals from sentinel-1," *SSRN Electronic Journal*, 2024. Available at SSRN: <https://ssrn.com/abstract=5222853>.
- [3] L. Kilic, C. Prigent, F. Aires, G. Heygster, V. Pellet, and C. Jimenez, "Ice concentration retrieval from the analysis of microwaves: A new methodology designed for the copernicus imaging microwave radiometer," *Remote Sensing*, vol. 12, no. 7, p. 1060, 2020. Presents the development of the IceCREAM algorithm for CIMR.